4. 직선운동과 회전운동: 모멘텀 해소의 두 가지 평형 형태

우주의 가장 근본적인 경향은 에너지 갭을 줄이고 디스턴스를 확장하는 것이다. 모든 에너지 아일랜드는 자신과 주변 사이에 존재하는 에너지 불균형을 해소하려는 방향성을 갖는다. 이 방향성은 우리가 일반적으로 '힘'이라고 부르는 것보다 더 근본적인 개념이며, DISTANCE에서는 이를 **모멘텀 해소 방향성**으로 해석한다.

가장 단순한 경우, 모멘텀 해소는 하나의 방향으로 이루어진다. 주변과의 에너지 갭이 특정 방향에서 가장 크게 감소할 수 있다면, 에너지 아일랜드는 그 방향으로 이동한다. 이것이 직선운동이다. 직선운동은 우주가 가장 직접적으로 에너지 갭을 줄이는 방식이며, 모든 방향의 디스턴스를 증가시키려는 우주의 거시적 경향과 가장 잘 부합하는 운동이다.

그러나 실제 우주에서는 대부분의 에너지 아일랜드가 독립적으로 존재하지 않는다. 하나의 아일랜드는 내부의 수많은 에너지 아일랜드들과 강하게 결속되어 있으며, 동시에 외부의 다른 아일랜드들과도 끊임없이 상호작용한다. 따라서 내부 모멘텀과 외부 모멘텀이 항상 동일한 방향을 요구하는 것은 아니다.

이처럼 여러 방향에서 서로 다른 모멘텀 해소 요구가 동시에 존재하면, 어느 한 방향으로 완전히 직선운동을 할 수 없게 된다. 이러한 상태에서 형성되는 운동이 바로 회전운동이다.

즉, 회전운동은 직선운동과 대립되는 개념이 아니라, **여러 방향의 모멘텀이 동시에 존재하는 환경에서 형성되는 평형 운동**이다. 이러한 의미에서 회전운동은 하나의 방향으로 완전히 해소되지 못한 직선운동이며, 다시 말해 **'실패한 직선운동'이 만들어낸 안정된 평형**이라고 해석할 수 있다.

회전은 흔히 각운동량이 보존된 결과로 설명된다. 물론 이러한 설명은 회전의 지속성을 기술하는데 매우 성공적이다. 그러나 그것은 이미 존재하는 회전을 어떻게 유지하는지를 설명할 뿐, **왜 회전이 시작되었는가**에 대해서는 충분한 답을 제공하지 않는다.

팽이는 스스로 돌지 않는다. 누군가 토크를 가해야 회전이 시작된다. 그렇다면 행성과 별은 누가 돌렸는가?

기존 천체물리학은 원시 성운 자체가 이미 회전하고 있었으며, 중력 수축 과정에서 각운동량이 보존되어 현재의 회전이 형성되었다고 설명한다. 그러나 이는 다시 다음 질문을 남긴다.

원시 성운은 왜 회전하고 있었는가?

DISTANCE는 이 질문을 다른 관점에서 접근한다.

초기 우주의 에너지 아일랜드들은 완전한 대칭 상태에 존재하지 않았다. 각 방향의 에너지 밀도와 에너지 갭은 서로 달랐으며, 모멘텀 해소의 방향 역시 공간적으로 균일하지 않았다. 이러한 비대칭 환경에서는 하나의 방향으로만 에너지를 해소하는 것이 항상 가능한 것은 아니다.

여러 방향의 모멘텀이 동시에 존재하고 서로 경쟁하는 과정에서, 직선적인 해소보다 회전이 더 안정적인 모멘텀 해소 방식이 되는 국소적 구조가 형성될 수 있다. 다시 말해, 회전은 외부에서 인위적으로 부여된 운동이 아니라, **불균형한 에너지 모멘텀이 스스로 만들어낸 자연스러운 평형 형태**이다.

초기의 미세한 회전은 이후 주변 에너지 아일랜드들과 지속적으로 상호작용하면서 변화한다. 주변의 에너지 밀도, 질량 분포, 거리, 그리고 다른 천체들의 모멘텀 흐름은 회전을 가속하거나 감속시키며, 장기간에 걸쳐 새로운 평형점을 향해 진화시킨다.

따라서 현재 행성의 자전은 단순히 형성 당시의 우연한 초기조건이 그대로 보존된 결과라기보다, 수십억 년 동안 주변 우주와 에너지 모멘텀을 교환한 끝에 도달한 **동적 평형 상태**로 해석할 수 있다.

이 관점에서 관성 역시 새로운 의미를 갖는다. 관성은 어떤 신비로운 힘이나 물질의 고유한 성질이 아니라, 기존 결속 구조 외부에 이를 변화시킬 새로운 모멘텀이 존재하지 않는 상태이다. 회전 역시 마찬가지이다. 현재의 회전은 어떤 내부 동력이 계속 공급되어 유지되는 것이 아니라, 주변 에너지 모멘텀과 이미 평형을 이루고 있기 때문에 안정적으로 지속되는 것이다.

결국 직선운동과 회전운동은 서로 다른 종류의 운동이 아니다.

직선운동은 하나의 방향으로 우세한 모멘텀 해소가 만들어낸 운동이며,

회전운동은 여러 방향의 모멘텀이 동시에 존재하는 환경에서 형성된 안정된 모멘텀 평형이다.

따라서 회전은 단순히 보존되는 운동이 아니라, **주변 우주와의 에너지 모멘텀 관계 속에서 끊임없이 형성되고 유지되는 평형 상태**이다.

이러한 해석은 행성과 별의 자전이 초기조건만으로 결정되는 것이 아니라, 질량, 크기, 공전거리, 중심별의 질량, 주변 천체와의 상호작용 등 우주적 모멘텀 구조 전체와 연결되어 있을 가능성을 제시한다. 만약 이러한 관계가 관측 데이터에서 검증된다면, 회전운동은 단순한 각운동량 보존의 결과가 아니라 우주의 에너지 평활화 과정이 만들어낸 보편적 평형 현상으로 이해될 수 있을 것이다.

4. Mass, Inertia, Motion, and Gravity

The Relational Mechanics of Bound Momentum

For centuries, gravity has been understood as an attractive force acting between masses. Isaac Newton described its magnitude with remarkable precision, while Albert Einstein reinterpreted the phenomenon as the curvature of spacetime. Yet both theories primarily describe how gravity behaves, rather than why it fundamentally arises.

DISTANCE does not regard gravity as an independent fundamental force. Instead, it interprets gravity as the macroscopic manifestation of nature’s universal tendency to reduce energy gaps.

However, to reinterpret gravity properly, we must first reconsider the concepts that stand before it: mass, inertia, and motion.

Gravity does not begin with attraction. It begins with bound momentum.

4.1 Mass as the Density of Bound Momentum

In conventional physics, mass is often understood as the amount of matter contained in an object, the measure of resistance to acceleration, or the source of gravitational interaction. These definitions are highly useful in calculation. They allow us to describe motion, predict orbits, and measure the strength of interaction between bodies.

But from the perspective of DISTANCE, mass is not simply the amount of material substance contained in an object. Nor is it merely a passive quantity that resists change.

Mass is the density of bound momentum within an energy island.

Every physical object is not a single indivisible entity. It is a collection of countless internal energy islands. Each of these internal islands possesses its own energy gap, its own momentum, and its own tendency toward smoothing. When these internal momenta are not dispersed outward independently but remain bound within a relatively stable structure, they form what we perceive as one object.

Mass is the degree to which these internal momenta are bound together while seeking to be smoothed into one coherent structure.

In this sense, mass is not merely "how much matter exists." It is the density of internal momentum binding. It expresses how strongly the momenta inside an island are held together as one independent structure.

A massive object is therefore not simply an object with more substance. It is an island whose internal momenta are more densely bound. Its internal energy islands are not isolated fragments. They are gathered, constrained, and held within a common structure of smoothing.

Mass is the internal density of this bound structure.

This definition allows us to understand mass not as an isolated material quantity, but as a structural condition of an island. It is the way internal momenta are gathered into a coherent unit.

Yet this does not mean that mass automatically produces an external physical effect.

If one island were ideally and absolutely isolated from all other islands, its internal momentum-binding density could still exist. Internally, the island could possess mass. But externally, that mass would have no physical meaning. It would not attract anything. It would not resist anything in relation to anything else. It would not appear as gravity, nor as measurable inertia from the perspective of another island.

Mass becomes physically effective only when a relation is formed.

Therefore, in DISTANCE, mass has two layers.

Internally, mass is the density of bound momentum within an island.

Externally, mass is the capacity of that internal binding to acquire physical effect through relation with another island.

A completely isolated mass is internally structured, but externally silent. It becomes a physical quantity with observable effect only when another island enters into a distance relation with it.

4.2 Inertia as the Ideal Absence of External Momentum Relation

The concept of inertia must also be redefined.

In conventional mechanics, inertia is often described as the tendency of an object to maintain its state of rest or uniform motion unless acted upon by an external force. This expression is useful at the level of observation. But it can easily give the impression that inertia is a kind of internal property or hidden force that makes an object resist change.

From the perspective of DISTANCE, inertia is neither a force nor a property.

Inertia is the ideal state in which an island has no momentum relation other than its existing internal binding structure.

If an island has no relation with any external momentum, then there is nothing outside it that can alter its relational condition. In that idealized state, the island remains only within its existing internal structure. This is what conventional physics describes as the maintenance of rest or uniform motion.

But this state does not exist in reality.

No island in the universe exists in absolute isolation. Even if the influence is extremely small, every island has some relation with other islands. There is always some distance relation, some momentum relation, and some smoothing relation. Therefore, inertia is not an independent physical force existing in nature. It is a limiting concept: the idealized absence of external momentum relation.

This interpretation is important because it prevents us from misunderstanding inertia as resistance.

Inertia is not the resistance of an object against change. What appears as resistance is the result of an external momentum relation being introduced into an already bound internal structure. When another relation enters, the existing structure is not instantly rewritten. The apparent delay, imbalance, or difficulty of change is not because inertia acts as a force. It is because the internal binding structure must be reconfigured through relation.

Thus, inertia is better understood as a reference condition, not a cause.

It is the hypothetical state in which only the internal binding of an island remains, without any external momentum relation.

This applies equally to rest, linear motion, and circular motion.

Rest is not absolute rest. It is only a state in which an island is relatively at rest with respect to another island or reference structure. In the universe as a whole, no island is completely motionless.

Likewise, motion is never absolute motion in empty space. It is always a change in distance relation.

Inertia is the ideal background against which such relational changes are interpreted.

4.3 Linear Motion as Directional Momentum Relation

If inertia is the ideal absence of external momentum relation, then motion begins when such a relation is formed.

Linear motion is the state in which an island forms an external momentum relation that appears to open in one dominant direction.

In ordinary language, we imagine an object moving through space along a straight line. But from the perspective of DISTANCE, the object is not simply passing through a pre-existing empty background called space. Rather, its distance relation with other islands is being reorganized in a particular direction.

Linear motion is therefore not motion through space. It is directional change in distance relation.

When an island is released into a relation where its momentum can be resolved in one dominant direction, its path appears linear. But this linearity is not absolute. It is a local approximation. At a certain scale, other curvatures and surrounding relations may be small enough to ignore, and the motion appears as a straight line.

In reality, however, no motion is completely free from other relations. Every island is surrounded by other islands, other gradients, other energy gaps, and other smoothing possibilities. Therefore, linear motion is not the pure form of motion in the universe. It is the simplified expression of a dominant relation.

This means that linear motion is not fundamentally different from other forms of motion. It is one way in which momentum seeks to resolve distance.

When the surrounding relational field does not significantly bend or constrain that direction, the motion appears straight.

Linear motion is thus the expression of momentum relation opened in one dominant direction.

4.4 Circular Motion as Bound Curvature

Circular motion appears very different from linear motion. A body does not move away in a straight line, but continues to curve around a center. Conventional mechanics explains this through centripetal force, orbital velocity, and the balance between inward acceleration and tangential motion.

DISTANCE does not reject these descriptions. But it interprets circular motion at a deeper relational level.

Circular motion occurs when linear smoothing does not proceed freely outward, but becomes constrained by the balance between internal momentum and external momentum.

An island tends to resolve its momentum through distance change. In relation to the larger universe, there is a global tendency toward the expansion or release of distance. However, a local island is not related only to the global field. It also contains internal islands, internal binding, and local momentum relations.

Because of these internal relations, the island cannot simply resolve all momentum by moving linearly outward. The internal momentum of the bound structure and the external momentum relation with other islands form a balance. In that balance, straight release is curved. The attempt at linear smoothing is captured within a local relational structure.

This is circular motion.

Circular motion is not merely movement around a center. It is the condition in which curvature is held within a bound relation.

The island does not simply fall inward. Nor does it simply escape outward. It remains in a sustained relational equilibrium between internal binding and external smoothing. Its path curves because the momentum that would otherwise move linearly is continuously redirected by relation.

In this sense, circular motion can be described as curvature trapped inside internal binding.

More precisely, circular motion is the state in which the momentum between internal islands and the momentum between external islands reach a form of equilibrium. The object continues to move, but its motion does not become simple escape. Its momentum is neither fully released nor fully collapsed. It is held in a curved relation.

Therefore, linear motion and circular motion are not separate mysteries. They are two different expressions of the same underlying tendency.

Linear motion appears when momentum relation opens in one dominant direction.

Circular motion appears when that directional tendency is constrained by internal and external momentum equilibrium.

In both cases, what we call motion is not the movement of a thing through empty space. It is the reconfiguration of distance relations among energy islands.

4.5 Gravity as the Directionality of Energy Smoothing

Only after mass, inertia, and motion are understood in this way can gravity be reinterpreted more precisely.

Gravity is usually described as attraction. A massive object is said to pull another object toward it. This language is natural because that is how the phenomenon appears. A stone falls toward Earth. The Moon remains in orbit around Earth. Planets move around the Sun. Everything seems to be drawn toward mass.

But DISTANCE does not interpret gravity as an independent pulling force.

Gravity is the directionality that emerges when internal momentum-binding density becomes relationally effective across distance.

Toward the center of a massive object, countless internal energy islands are densely bound. This direction therefore provides a much shorter average energy distance than other directions. It also offers more pathways through which energy gaps can be reduced.

From the perspective of another nearby island, not every direction offers the same opportunity for energy smoothing. Some directions contain relatively few bound structures. Other directions contain a much higher density of internally bound momentum. The direction toward a massive object is special because it contains greater smoothing possibility.

This is what appears macroscopically as gravity.

The object is not pulled because mass emits a mysterious attractive agency. Rather, it moves toward the direction where the relational structure of energy smoothing is more strongly formed.

The center of a massive body does not pull in the psychological or mechanical sense of attraction. It provides, in relation, a denser field of smoothing possibility. From the perspective of the nearby object, the direction toward the massive body is the direction in which energy distance is shorter, smoothing pathways are denser, and momentum can be more effectively resolved.

This is why falling occurs.

Falling is not the result of a thing being grabbed by gravity. It is the movement of an island along the direction in which its relation to other bound momentum structures allows energy gaps to be smoothed more effectively.

Thus, gravity is not the cause that externally commands motion. Gravity is the name we give to the relational direction in which energy gaps become easier to smooth.

4.6 Universal Gravitation as the Quantitative Form of Relational Smoothing

Newton's law of universal gravitation remains one of the most powerful quantitative descriptions in the history of science:

$$
F = G \frac{m_1 m_2}{r^2}
$$

DISTANCE does not reject this formula. It does not attempt to replace its quantitative structure. Instead, it reinterprets the meaning of the terms within it.

In conventional interpretation, (m_1) and (m_2) are the masses of two objects, and (r) is the distance between them. The gravitational force increases with the product of the two masses and decreases with the square of the distance.

In DISTANCE, (m_1) and (m_2) are interpreted as the internal momentum-binding densities of two islands.

The term (m_1m_2) expresses the relational capacity of two internally bound structures to participate in mutual smoothing. The greater the internal momentum-binding density of each island, the greater the relational possibility for smoothing between them.

But this does not mean that either mass produces gravity by itself.

A mass in absolute isolation has no external gravitational effect. Its internal binding may exist, but it has no physical efficacy outside itself. Gravitational effect appears only when another island enters into relation with it.

Therefore, (m_1m_2) does not mean that two independent substances mysteriously pull each other. It means that two internal momentum-binding structures become mutually effective when a distance relation is formed.

The term (r^2) expresses the dispersion of that relational possibility across external distance. As the distance between two islands increases, the smoothing relation is distributed over a larger relational field. The effective directionality therefore weakens in proportion to the square of distance.

In this way, Newton's formula remains valid as a quantitative expression, while its ontological meaning is reinterpreted.

It is not that two masses attract each other through an independent force acting across empty space. Rather, two internally bound momentum structures form a relational smoothing direction whose effective strength is proportional to the product of their internal binding densities and inversely proportional to the square of their external distance.

Newton described the measurable structure of gravitational behavior.

DISTANCE seeks to reinterpret why that structure appears.

Universal gravitation is the quantitative expression of how two internally bound momentum structures acquire relational effect across distance.

4.7 From Force to Relation

The central shift of this interpretation is simple but fundamental.

Gravity is not an isolated force.

Mass is not an isolated substance.

Inertia is not a hidden property of resistance.

Motion is not the travel of a thing through empty space.

All of these are relational expressions of bound momentum and energy smoothing.

Mass is the density of bound momentum within an island.

Inertia is the ideal absence of external momentum relation.

Linear motion is the opening of momentum relation in one dominant direction.

Circular motion is the confinement of curvature within the equilibrium of internal and external momentum.

Gravity is the relational directionality of energy smoothing.

Universal gravitation is the quantitative form of that relational directionality.

What exists first is not force, but difference. What follows from difference is momentum. What momentum seeks is smoothing. What appears through relation is motion. And when internally bound momentum structures encounter one another across distance, the directionality of that smoothing appears as gravity.

Therefore, gravity is not the cause of falling.

Gravity is the relational direction in which falling becomes possible.

Mass is not the independent source of attraction.

Mass is the density of bound momentum that becomes physically effective only through relation.

The universe is not filled with isolated things pulling one another across empty space. It is filled with islands of bound momentum, each participating in the continuous smoothing of distance.